

ECON 3040 - Chapter 3: Statistics Review

Ryan T. Godwin

University of Manitoba

What is a *statistic*?

Examples:

- ▶ 9 out of 10 dentists choose Colgate
- ▶ The richest 1% own half the wealth
- ▶ Surgeons who played more than 3 hours / week of video games had 37% fewer errors
- ▶ Vladimir Guerrero Jr. hit his first home game home run during the 150th game in the 2026 season
- ▶ The maximum mark in ECON 3040 2025 A01 was 99.6%
- ▶ The median after tax income of Canadian families was \$75,500 in 2024
- ▶ The average age of retirement in Canada is 65.4 years

How were these statistics calculated?

Statistics Review

- ▶ A **statistic** is a function of a sample of data
- ▶ An **estimator** is a statistic
- ▶ **Population parameter** $\rightarrow$ unknown
- ▶ **Estimator** $\rightarrow$ used to estimate an unknown population parameter
- ▶ The sample, y , will be considered **random**
- ▶ Since y is random, estimators using y will be **random**

Since estimators are random, they have a _____, given a special name: **sampling distribution**.

We will obtain properties of the sampling distribution to see if the estimator is “good” or not.

3.1 Random Sampling from the Population

- ▶ Typically, we want to know something about a population
- ▶ The population is considered to be very large (infinite), and contains some unknown “truth”
- ▶ We likely won’t observe the whole population, but a sample from the pop.
- ▶ We’ll use the sample, y , to estimate that something

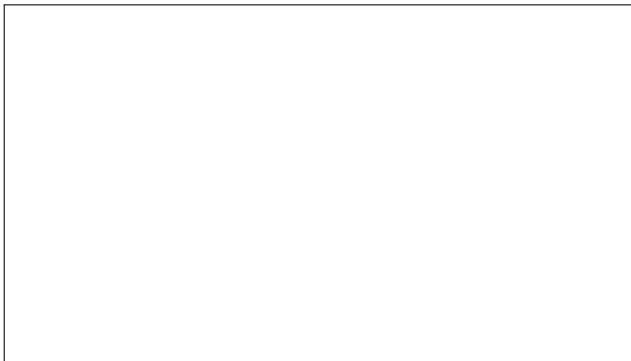
Example: Mean Height of a U of M Student

Suppose we want to know the mean height of a U of M student.

Let y = height of a student.

- ▶ Population: all students
- ▶ Population parameter of interest: μ_Y

We can't afford to observe the whole pop. We'll have to collect a sample, y .



How Should the Sample Be Selected?

We want the sample to reflect the population.

Question: How should the sample be selected from the population?

In particular we want the sample to be i.i.d.

- ▶ **Identically**
- ▶ **Independently**
- ▶ **Distributed**

The Sample Is Random

So, the sample y is random!!

- ▶ Could have gotten a different y
- ▶ Parallel universe